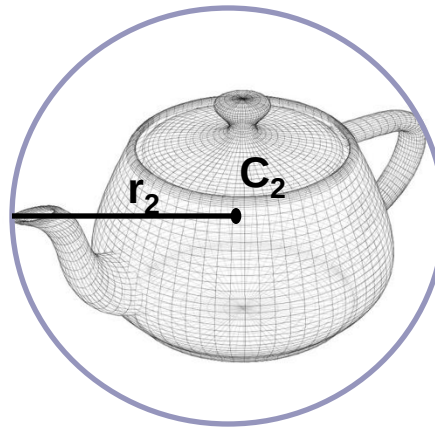
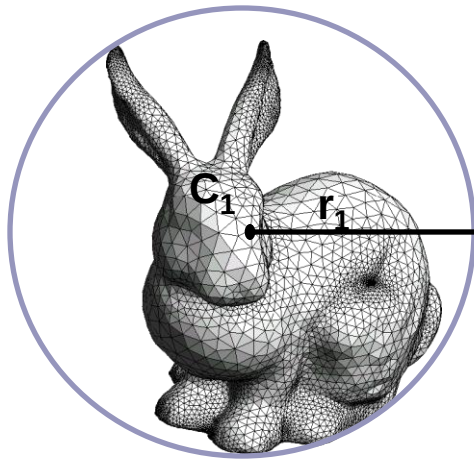


Collision Detection

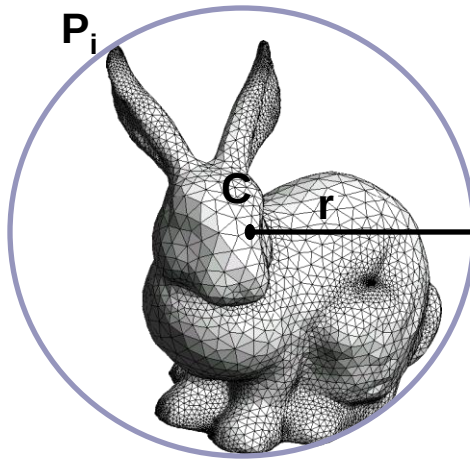
Bounding Spheres

- Place **spheres** around objects
- If spheres do not intersect, objects do not intersect
- Spheres intersect if $\|C_1 - C_2\| < r_1 + r_2$
- Easy



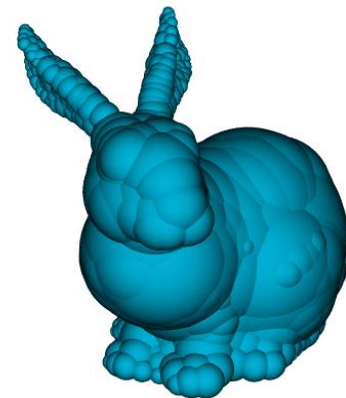
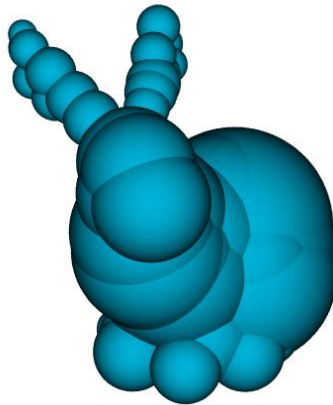
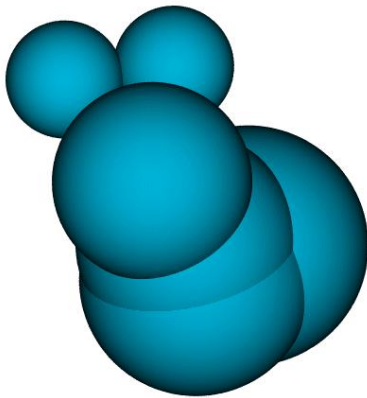
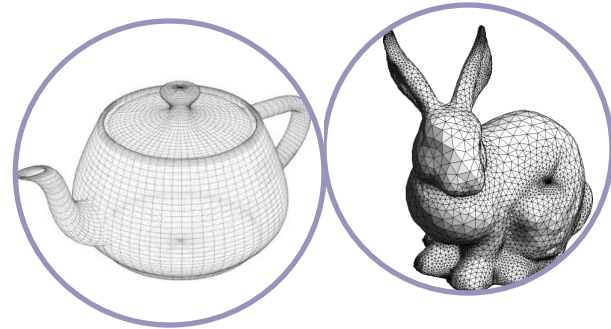
Bounding Spheres

- center = $((x_{\min} + x_{\max})/2, (y_{\min} + y_{\max})/2, (z_{\min} + z_{\max})/2)$
- radius = $\max_i \|C - P_i\|$



Bounding Spheres

- Hierarchical Sphere Tree



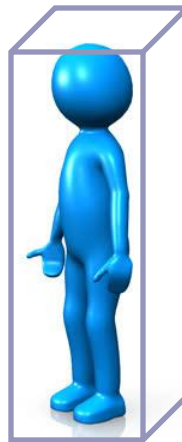
Bounding Cylinder

- Place **cylinder** around objects around tall, thin objects
- Cylinders intersect if $\|C_1 - C_2\| < r_1 + r_2$



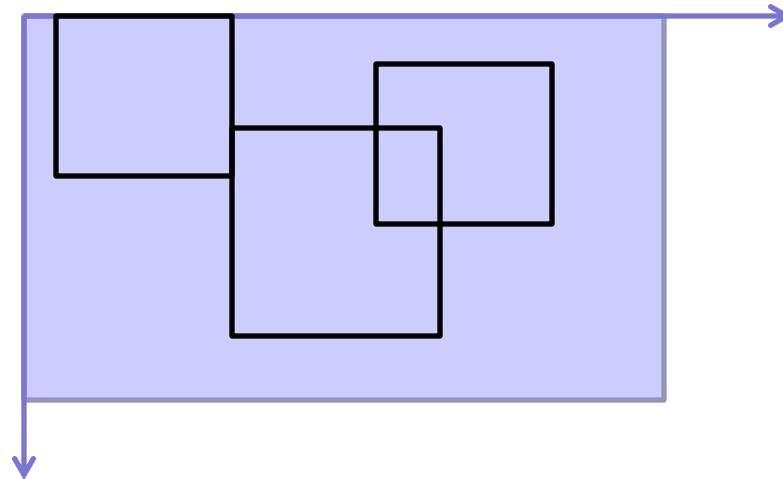
Bounding Boxes

- Place **a box** around objects
- For testing against floor, walls, ceiling
- Two types of bounding boxes can be used:
 - axis-aligned
 - freeform



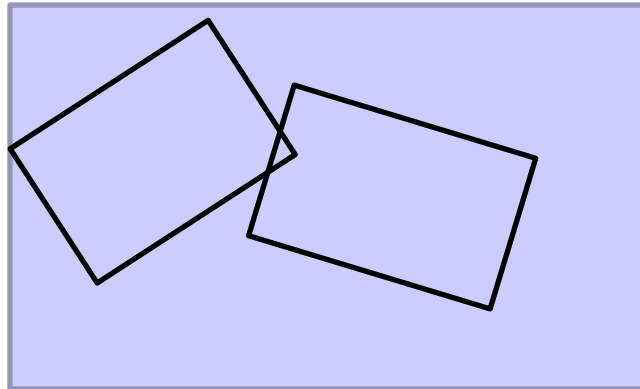
Bounding Boxes

- Axis-aligned bounding boxes



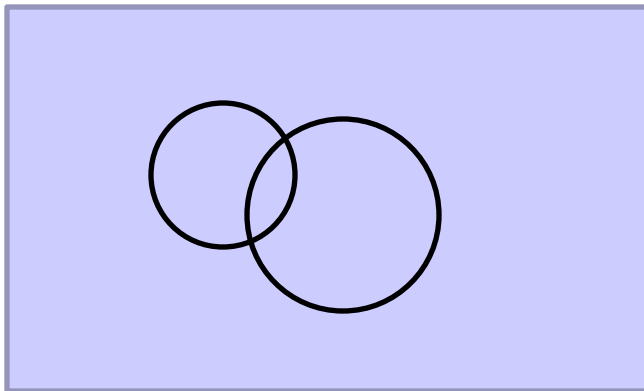
Bounding Boxes

- Freeform bounding boxes

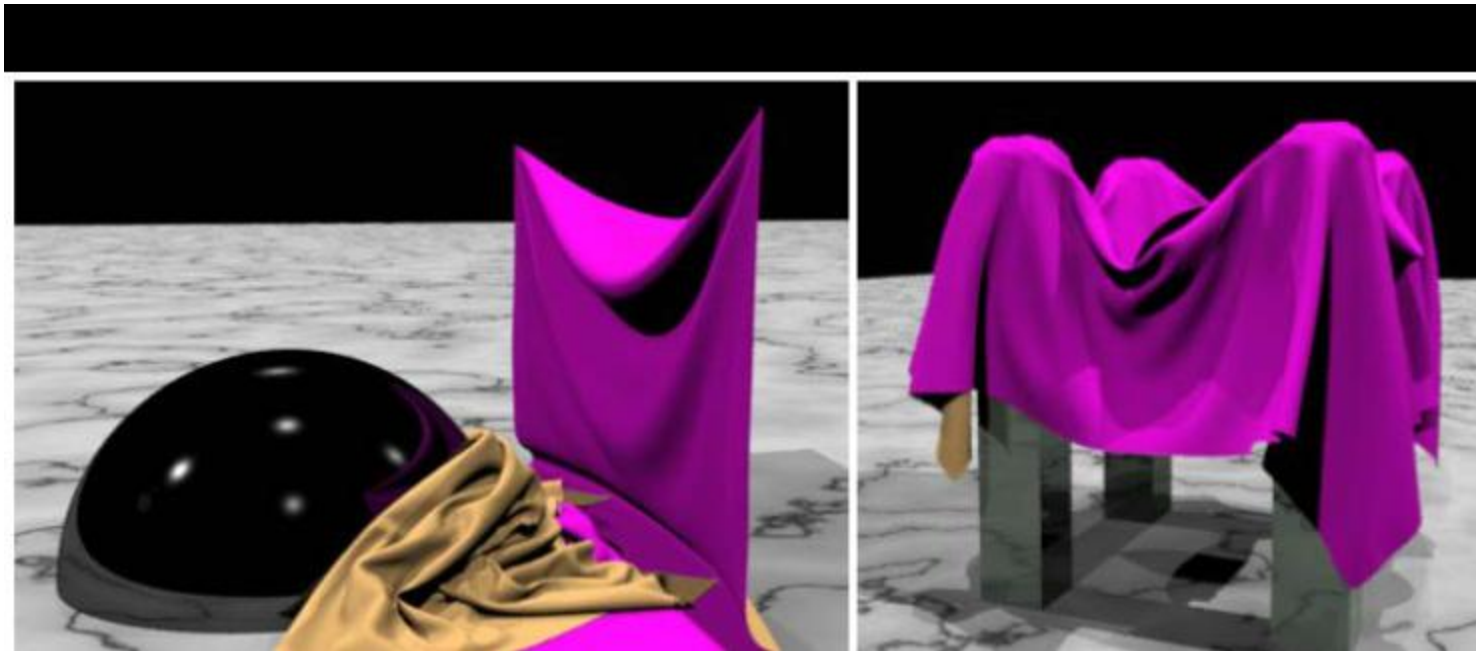


Object-to-Object Sliding

- Previously, if an object-to-object collision occurred, the object must be moved to its original location
- The moving object slides to the side of the static object.



Cloth Collision



Cloth Collision

Particle collision

- <https://www.youtube.com/watch?v=ROOoj9t6-NA>
- <https://www.math.ucla.edu/~jteran/>